

MEET THE CULPRIT

CITRUS ARRESTED FOR GRAVE OFFENSE AGAINST HUMANITY?



LIMONENE

40%

Whats your common name?

Limonene

Chemical Formula

C₁₀H₁₆

136.2340

Molar Weight

Add yours

1-methyl-4-(prop-1-en-2-yl)cyclohex-1-ene



COMPLAINT

I CAN'T SEE
It That
WELL

Polarity: **Nonpolar**

Melting point:

≈ **-74 °C**

Water solubility: **Very low**

Boiling point:

≈ **176 °C**

Odor: **Strong, characteristic citrus scent**

Appearance:

Colorless to pale-yellow liquid

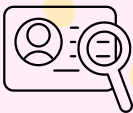
MORE

KEY STRUCTURAL FEATURES

TWO
C=C DOUBLE BONDS

SIX-MEMBERED
CARBON RING

NON-POLAR
HYDROCARBON



WHERE IS IT USED?

used as a citrus
flavoring ingredient.

FLAVOR



can dissolve oily
materials; useful in
some cleaners and
degreasers.

**CLEANING
PRODUCTS**

used as a
fragrance
ingredient

COSMETICS



EXTRACTION

Recover essential oil from
citrus peel

SEPARATION

Separate the oil-rich
mixture into its components

PURIFICATION

Obtain limonene with the
desired purity

FORMULATION

Incorporate the purified product
into commercial products.

TO ENGINEERING

A 2021 study found that about half of the identified VOCs in urban air came from chemical products, including limonene.

When released into the atmosphere, it can participate in reactions that contribute to ground-level ozone formation

★★★★★

TESTIMONIAL

WHAT DID I DO?



"NOTORIOUS
OZONE
GENERATORS" DID YOU
I'M AMAZING!!! KNOW?

how does
IT BEHAVE?

HANDLE WITH CARE! IRRITANT
FLAMMAABLE
SKIN SENSITIZER
ENVIRONMENTAL HAZZARD

CONNECTION

1

American Chemical Society. (2020). Limonene.

<https://www.acs.org/molecule-of-the-week/archive/l/limonene.html>

2

NIST Chemistry Webbook. (n.d.). Limonene.

<https://webbook.nist.gov/cgi/cbook.cgi?ID=138-86-3>